Who Delivers When It Matters?

A pressure-performance rating for IPL and T20I batters and bowlers

SupeRR Selector Season 2 | Round 1 Methodology Report

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Data: 707 matches, 168,596 deliveries (IPL 2022-2026 and T20Is from 2022)

Summary of the approach

We built a ball-by-ball **Pressure Index** (0-100) from the match situation, flagged the top quarter of balls as high-pressure, and compared every batter and bowler with what an average player produces in the same situation. Each player's pressure rating starts from their own usual level and is adjusted by what they actually did under pressure, so a handful of lucky or unlucky balls cannot decide a ranking. The final score combines the rating (65%) with consistency across matches (35%).

The final top-10 lists are on the next page. We also report how firm each rank is. The honest headline: pressure skill is hard to separate from luck in a four-year sample, so the lists show the **best-supported** players, and the report quantifies the uncertainty rather than hiding it.

1. Final rankings

These are the lists entered on Unstop, in rank order. "Top-10 chance" is the share of 300 bootstrap re-samples of matches in which the player stayed in the top 10.

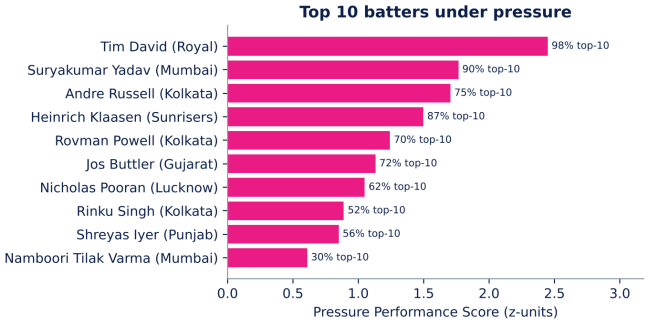
Top 10 batters under pressure

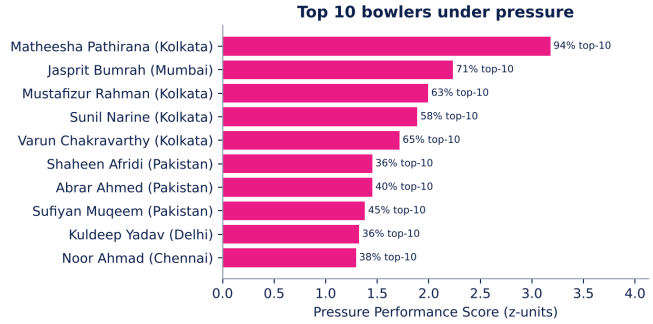
Rank	Player	Team	Score	Rating*	Consistency	Pressure balls	Top-10 chance
1	**Tim David**	Royal Challengers Bengaluru	2.45	+24.9	69%	363	98%
2	**Suryakumar Yadav**	Mumbai Indians	1.76	+19.1	70%	315	90%
3	**Andre Russell**	Kolkata Knight Riders	1.70	+19.2	68%	251	75%
4	**Heinrich Klaasen**	Sunrisers Hyderabad	1.50	+19.4	63%	320	87%
5	**Rovman Powell**	Kolkata Knight Riders	1.24	+16.6	65%	508	70%
6	**Jos Buttler**	Gujarat Titans	1.13	+15.1	67%	186	72%
7	**Nicholas Pooran**	Lucknow Super Giants	1.05	+15.9	62%	322	62%
8	**Rinku Singh**	Kolkata Knight Riders	0.89	+16.1	59%	386	52%
9	**Shreyas Iyer**	Punjab Kings	0.85	+14.8	61%	227	56%
10	**Namboori Tilak Varma**	Mumbai Indians	0.61	+11.5	65%	545	30%

Top 10 bowlers under pressure

Rank	Player	Team	Score	Rating*	Consistency	Pressure balls	Top-10 chance
1	**Matheesha Pathirana**	Kolkata Knight Riders	3.18	+17.0	75%	497	94%
2	**Jasprit Bumrah**	Mumbai Indians	2.23	+14.6	61%	443	71%
3	**Mustafizur Rahman**	Kolkata Knight Riders	1.99	+11.8	66%	434	63%
4	**Sunil Narine**	Kolkata Knight Riders	1.89	+11.6	64%	361	58%
5	**Varun Chakravarthy**	Kolkata Knight Riders	1.72	+10.6	64%	593	65%
6	**Shaheen Afridi**	Pakistan	1.45	+8.9	64%	264	36%
7	**Abrar Ahmed**	Pakistan	1.45	+7.0	71%	269	40%
8	**Sufiyan Muqeem**	Pakistan	1.38	+7.2	69%	151	45%
9	**Kuldeep Yadav**	Delhi Capitals	1.32	+9.3	59%	454	36%
10	**Noor Ahmad**	Chennai Super Kings	1.29	+12.0	47%	399	38%

*Rating = estimated runs per 100 pressure balls above the average player in the same situation (wickets counted as lost runs). Consistency = share of pressure innings in which the player's net contribution was positive.





2. How we defined pressure

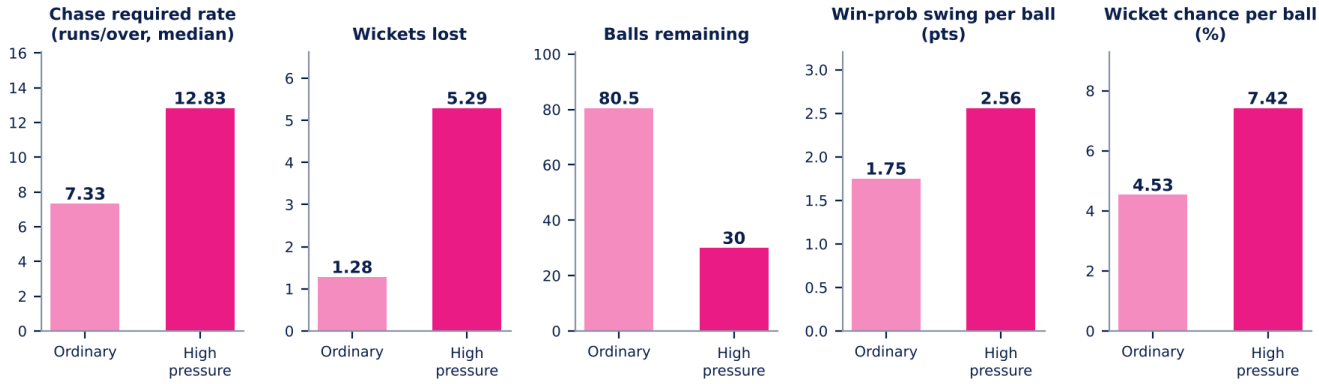
Pressure is not one number on the scoreboard. A team needing 12 an over with 8 wickets in hand in over 8 is relaxed; the same rate with 3 wickets left in over 17 is not. We therefore combine four signals into a Pressure Index (PI) for every delivery:

- **Stage (S):** how much of the innings is gone, S = ((balls bowled + 1) / 120)$^{0.75}$. Early balls can't decide a match; late balls can.
- **Closeness (C):** how balanced the contest is, C = 1 - |2 x WP - 1|, where WP is the batting side's win probability from a gradient-boosted model trained on match states (balls left, wickets, runs, target, required and current rate, competition). Out-of-fold AUC is 0.68 for the 1st innings and 0.89 for chases.
- **Squeeze (Q) in a chase:** 0.65 x required-rate strain + 0.35 x wicket strain. Rate strain = clip((required rate - 6) / 10, 0, 1); wicket strain = clip((wickets lost - 2) / 6, 0, 1).
- **Squeeze (Q) in the 1st innings:** 0.5 x how far the score is behind the typical score at that ball (per 35 runs) + 0.5 x wicket strain. A first innings is under pressure when the team is in trouble.

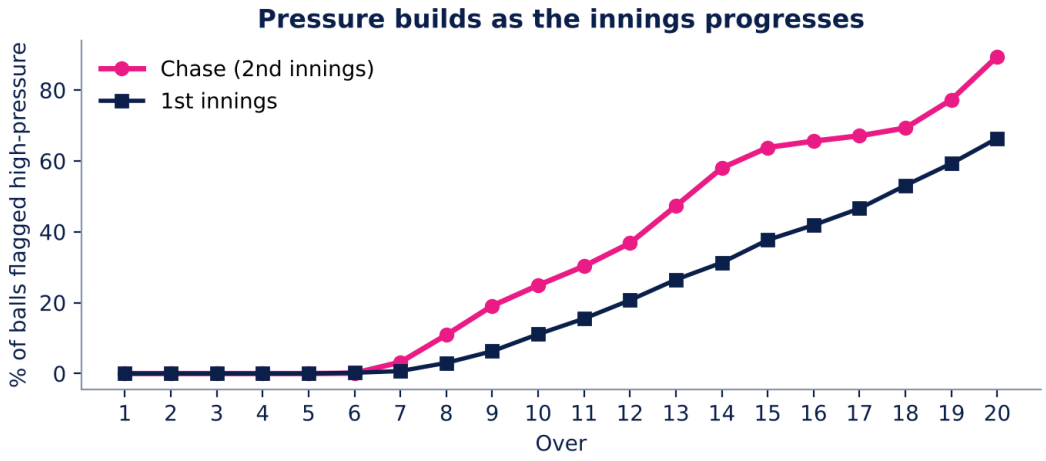
PI = 100 x S x (0.5 x C + 0.5 x Q) in a chase, and **PI = 100 x S x Q** in the first innings (closeness is uninformative before a target exists).

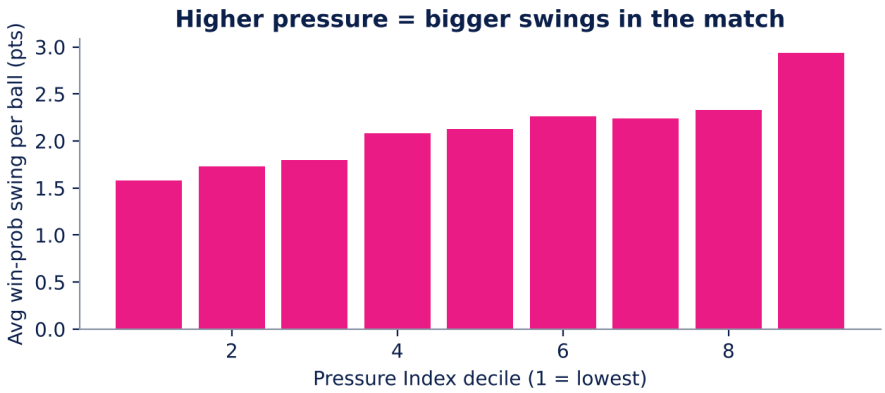
High-pressure ball: PI of at least 26.0 (top 25%, 42,149 balls). **Ordinary ball:** PI of at most 10.5 (bottom 50%). These cut-offs were fixed before ranking players.

Why this separates pressure from ordinary play



High-pressure balls come with a chase median required rate of 12.83 against 7.33, 5.29 wickets lost on average against 1.28, and 30 balls left against 80. Each ball moves the win probability by 2.56 points on average against 1.75, and a wicket falls on 7.42% of balls against 4.53%.





The index was built from match state only, never from outcomes of the ball itself, yet the average swing in win probability rises across its deciles. That is the independent check that it measures real stakes.

3. How we measured performance

Run value of a ball

Every delivery gets a **run value**: runs scored (including extras) minus a wicket cost, minus what an average player produces in the same situation. The wicket cost is the number of runs a wicket is worth, estimated by regressing the change in win probability on runs and wickets: 10.9 / 9.3 / 7.2 (1st innings: powerplay / middle / death) and 7.3 / 6.1 / 3.4 (chase). The "same situation" is defined by competition type (IPL or T20I), innings, over, wickets lost (0-2, 3-5, 6+) and, for batters, batting-order group (1-3, 4-5, 6-7, 8+) or, for bowlers, pace or spin. This stops finishers from being compared with openers, and death-over bowlers with powerplay bowlers. A bowler's value is the same number with the sign reversed.

Player baseline and pressure rating

For every player we compute the average run value on ordinary balls (their usual level) and on high-pressure balls. Because 120-300 pressure balls is a small sample, we use an **empirical-Bayes** estimate: the pressure rating starts from what the player's usual level predicts, then moves toward the observed pressure value in proportion to how many balls we have. Between-player spread was estimated from the data and equals about 1,196 balls (batters) and 1,216 balls (bowlers) of prior weight. In plain terms: a player needs a long record before their pressure results outweigh their general quality.

Consistency

We count each match in which the player faced or bowled at least 6 high-pressure balls and mark it positive if the net run value was above zero. Consistency = (positives + 3) / (matches + 6), which pulls small samples toward 50%.

Final score

Score = 0.65 x z(pressure rating) + 0.35 x z(consistency), where z is the standardised value among eligible players. The brief asks for players who deliver *consistently*, which is why consistency carries a third of the weight.

A clutch-shift test, and why it is not scored

We also tested the gap between each player's pressure level and their usual level. For batters the estimated between-player variance of this gap is indistinguishable from zero, and for bowlers it is small. In other words, the data do not show players who are reliably better or worse under pressure than their general quality predicts; they show players whose general quality is high and whose pressure results hold up. We therefore report the pressure and baseline figures side by side in the dashboard but give the shift no weight. Adding it back at 25% keeps 9 of 10 batters and 8 of 10 bowlers unchanged.

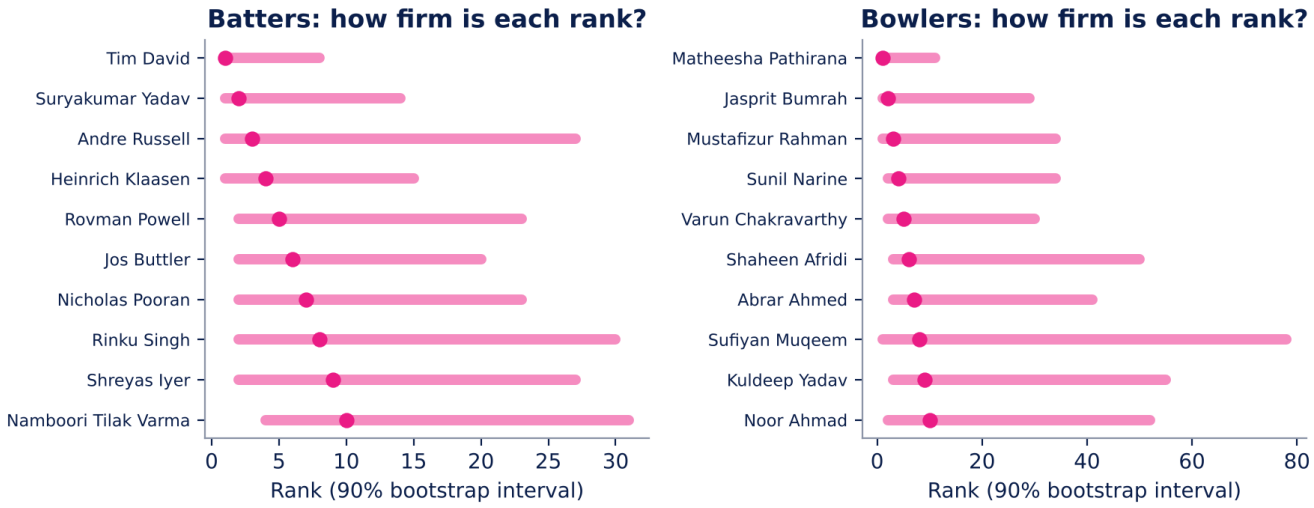
Supporting metrics (not in the score)

For readability the dashboard also shows classic numbers on pressure balls: strike rate, boundary %, dot % and dismissals for batters; economy, dot %, boundaries and wickets for bowlers.

4. Fair comparisons

- **Sample size:** a player must have at least 120 high-pressure balls, at least 10 high-pressure appearances (6+ balls each) and at least 60 ordinary balls so that a baseline exists. This leaves 45 batters and 108 bowlers. Everyone else is not ranked.
- **IPL and T20I together:** the situation baseline is split by competition type, so a run in the higher-scoring IPL is compared with other IPL runs. Players who appear in both are pooled; the match-state model also uses competition type as an input.
- **Opposition strength:** balls against associate nations are given half weight; balls against full members and IPL teams get full weight.
- **Match context:** expected values depend on over, wickets lost and role, so players are never credited for simply batting at the death or bowling in the powerplay.
- **Data hygiene:** we used only IPL 2022-2026 and T20I matches. Asia Cup Rising Stars and the Emerging Teams Asia Cup (A-team events) are excluded. Super overs, abandoned or no-result matches and rain-revised matches are excluded because the target is not comparable. Wides and no-balls do not count as balls faced or bowled; non-striker run-outs are not charged to the striker; run-outs are not credited to the bowler.

Evidence for the top 10



Each bar shows the middle 90% of ranks the player received across 300 re-samples of matches. Narrow bars (for example Tim David and Matheesha Pathirana) are well supported. Wide bars mean the player is in the group of contenders but the exact order is uncertain.

Why the leaders lead

Batters. Tim David scored 641 runs off 363 high-pressure balls (strike rate 176.6, 21 dismissals) and remained clear of the field after adjusting for role and situation. The group at the top is dominated by middle-order and finishing batters because high pressure is concentrated in the second half of an innings; openers rarely face enough high-pressure balls to qualify.

Bowlers. Matheesha Pathirana conceded 504 runs from 440 high-pressure balls (economy 6.87, 40 wickets, 36.4% dot balls) and has the highest consistency among the top 10 (75% of pressure matches positive).

5. Reliability and limitations

Checks we ran

Check	Batters	Bowlers
Top-10 overlap when only the pressure rating is used (no consistency)	9/10	8/10
Top-10 overlap with 50/50 rating and consistency weights	9/10	8/10
Top-10 overlap with 80/20 weights	9/10	10/10
Top-10 overlap when pressure = top 30% of balls	7/10	7/10
Top-10 overlap when pressure = top 20% of balls	8/10	8/10
Split-half correlation of the final score (two random halves of matches, 30 repeats)	0.15	0.07

What the checks say

- The lists are **robust to how we set the weights and the pressure threshold**: at least 7 of 10 names survive every change, and 8-10 of 10 survive most.
- The split-half correlation is low. Two random halves of the matches give only weakly similar orderings. This is a known property of pressure statistics: a player sees roughly 150-550 high-pressure balls in four years, and ball-level outcomes are very noisy. We chose a shrinkage method and a minimum sample precisely because of this, but nobody should read the exact order within the top 10 as settled.
- Because the rating borrows heavily from each player's usual level, it rewards players who are good in general *and* do not fall away under pressure. It cannot, on this data, identify players who are specifically better in the moment.

Limitations

- Bye and leg-bye runs are not recorded per ball in the file, so they are counted as 1 run each; the first-innings total matches the stated margin exactly in only about 30% of decided-by-runs matches, so a small amount of noise remains in the match state.
- The win-probability model has no team-strength or venue input, which makes first-innings win probability close to 50/50 and is why first-innings pressure is defined through trouble (behind par, wickets lost) instead of closeness.
- Opposition strength is a two-level weight, not a full rating. T20I players who face weaker sides may be slightly flattered; franchise players face a consistently strong pool.
- Players with fewer than the minimum balls (for example, many openers and part-time bowlers) are not ranked, even if their few pressure balls were excellent.
- Four seasons of data and one hold-out-free sample: we have not tested the ranking on data the model never saw.

Reproducibility

All steps use the supplied CSV files only: filter matches, build ball-level match states, fit the win-probability model, compute the Pressure Index, compute run values with situation baselines, apply the shrinkage and consistency, and rank. Code is supplied in *superr_pipeline.py*; ratings for every eligible player are in the accompanying CSV files.